

How Precision Affects User Trust in Teleoperation of Robot Arm using Keyboard

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Figure 1: The setup of our experiment

Abstract

This study examines how humans perceive and trust robots that exhibit different random behaviour under uncertainty. We conducted a user-study with 22 participants, where participants collaborated with a Panda robot arm on a pick-and-place task. In the study, two types of robot behaviour were compared. The first group operates the robot according to the commands entered by the subject. Meanwhile, participants in another group operate a robot that performs random movements with a certain probability, differing from the commands entered by the target. The result showed that when robots performed random movements contrary to the subjects' intentions, their trust in the robots decreased compared to the group that operated robots that moved as intended. It was also found

that robots exhibiting random movements imparted a sense of unpredictability to the subjects. However, there were also instances where subjects did not perceive these as errors.

These findings indicate that the system's human trust in robots depends on the system's unpredictability and random behaviour. Even if subjects don't recognize it as a system error, robots that are difficult to predict will struggle to gain human trust. This study highlights the importance of robot predictability in human-robot interaction.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Keyboards*; *User studies*.

Keywords

Teleoperation, Human-Robot Interaction, Robot Arm, Franka Panda, Trust, Predictability

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1 Introduction

Human-Robot interaction (HRI) in real-world environments inherently involves uncertainty, which introduces random unpredictable behaviour of robots during interaction. Understanding how humans perceive and respond to randomness is essential. Trust in robots depends not only on objective performance but also on perceived randomness and uncertainty.

Previous HRI research has shown the relationship between readability and predictability of robot behavior and psychological impact. However, existing studies largely focus on overall performance or reliability framing, rather than treating randomness itself as the primary cause. Much existing research focuses primarily on comparing good design (safety and human-centered considerations) with bad design (performance deficiencies and malfunctions), and does not directly address how “intentionally unpredictable randomness in behavior” is perceived by humans. Furthermore, the impact of unpredictable movements in situations where humans operate or collaborate with robots has not been sufficiently verified.

To address this gap, we investigate how robot performance (with randomness or without randomness) influences human trust and perception. This study focuses on a task where participants manipulate a robotic arm to complete the pick-and-place for each block in sequential order. It examines the perceptual difference between the group who used a robot, randomly executing movements differing from the subject’s intention and another group who used a robot that operates precisely according to the subject’s intent.

This study focuses on a task where subjects use a robotic arm to sequentially move multiple blocks to designated positions. It examines the perceptual difference between a robot that randomly executes movements differing from the human’s intended actions and a robot that operates precisely according to the subject’s intent. By exploring this question, this work aims to demonstrate that the randomness and predictability of a robot’s behavior significantly influence human trust. In addition to that, it’s emphasizing the importance of incorporating human-centered considerations into robot planning and control design.

2 Study Design

We used a between group experiment method where we divide participants into two groups and each group experiences different conditions: with randomness of the movement of the robot arm or no randomness.

- (1) In this study, individual variables were set as whether randomness is incorporated into the robot’s movements, then the feedback received regarding user trust and mental states was treated as the dependent variable.
- (2) The hypothesis was formulated prior to the experiment. This study is designed to examine how accuracy of the robot movement affects the human trust towards the robot. The hypothesis tested in this study was,

H1: Accurate and predictable robot movements will result in higher perceived trust compared to inaccurate robot movements.

As a manipulation check, we measured the baseline trust levels towards the robot from participants by using questionnaires. This ensured that there were no significant differences in initial trust between participant groups before the task. To evaluate how the robot’s behaviour affects the subject’s trust and interaction behaviour, we conducted objective and subjective measures. The participant’s subjective perception is measured by questionnaires that we conducted after the experiment. The questionnaires include the questions that explore participants’ trust in the robot as well as their perceived psychological workload during the interaction. In addition to the subjective measure, several metric objective measures were conducted.

- Time to complete the task
- Number of random movement occurred
- Direction of the random movement
- Joint state of the arm
- At what time, which button was pressed
- List of all keyboard inputs from participants

These measures were used to analyze participants’ interaction strategies and behavioral responses to different robot movement conditions.

3 Task Design

3.1 Task Description

This study investigated how participants respond to a robot’s movement accuracy in a claw machine task. The participants controlled a Panda collaborative robot arm to collect and discard physical blocks placed on a 12×18 grid, where each cell measured 3 cm × 3 cm as shown *below*. The task required discrete 2D planar movements (North, South, East, West) with 3 cm increments to locate and retrieve blocks, followed by predetermined movement to a discard pile.

The experimental setup consisted of a printed grid with eight colored blocks numbered 1–8 placed at strategic positions. Block 4 (orange) served as the home position, where the robot arm returned after each block was successfully discarded. The block positions were predetermined to ensure that participants would be required to navigate in all cardinal directions during the task. This setup

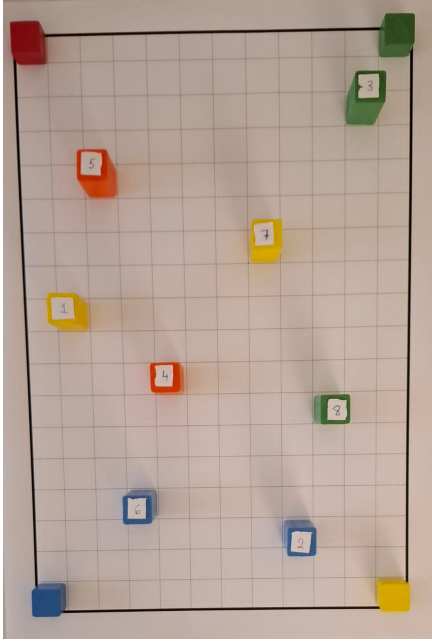


Figure 2: Setup of sequence for claw machine task

ensured that all participants performed the task under identical conditions.

3.2 Implementation

Participants controlled the robot arm using keyboard input (W, A, S, D keys for directional movement and C key to trigger the pick-up sequence). When the PICK command was executed, the robot performed the following automated sequence: descended 6 cm vertically, closed the gripper to secure the block, ascended 30 cm to clear the work surface, calculated and executed a trajectory to the discard pile (visualized in RViz), positioned above the discard pile, opened the gripper to release the block, and returned to the home pose defined by fixed joint states. The task was completed when all eight blocks were successfully transferred to the discard pile.

Two experimental conditions were implemented to test the hypothesis. Group B (Control – Normal Movements): The robot executed all directional commands precisely as input, moving exclusively in cardinal directions (North, East, South, West) with 3 cm increments. Group A (Experimental – Noisy Movements): The robot received corrupted movement commands designed to simulate inaccuracy. A random number (1–4) determined the interval between noisy commands. When a noisy movement was scheduled, a random number (1–100) was generated: if the number exceeded 90, the movement executed normally; otherwise, the movement was transformed into a diagonal direction based on the remainder when divided by 4 (Remainder 1: North-West, Remainder 2: North-East, Remainder 3: South-East, Remainder 0: South-West). This mechanism ensured that movements were unpredictable to the participant, as they could not anticipate when a noisy movement would occur or in which diagonal direction the robot would deviate.

Table 1: Questionnaire responses

Participant ID	Used similar systems	Careful with unfamiliar automation	Trust system rather than mistrust
242	4	7	7
616	4	5	4

4 Method

4.1 Participants

A total of 21 participants took part in the study, recruited from a university community. No specific inclusion or exclusion criteria were applied beyond the ability to provide informed consent. Participants were randomly assigned to either Group A (Noisy Movements, $n=11$) or Group B (Normal Movements, $n=10$), with alternating assignment to distribute conditions evenly across the recruitment sequence.

4.2 Procedure

Upon arrival at the laboratory, participants were informed about the study and provided with a consent form, which they reviewed and signed. Each participant then completed a pre-study questionnaire designed to establish baseline measures of familiarity with automated systems and general propensity to trust technology.

The experimenter provided a detailed explanation of the claw machine task, including the control scheme (W, A, S, D for movement; C for pick-up), the objective (retrieve all eight blocks and place them in the discard pile), and safety considerations. Participants were not informed about the presence or nature of the experimental manipulation (movement noise). Each participant was assigned a randomly generated ID number, which was recorded in both the log files (ROS bags) and the questionnaire responses to maintain anonymity while allowing data linkage.

Participants then performed the teleoperation task, controlling the robot arm to sequentially pick up each of the eight blocks and place them in the discard pile. The task typically required approximately 5 minutes to complete. Upon task completion, the participant pressed the Q key to terminate the teleoperation session.

Finally, participants completed a post-study questionnaire designed to assess their subjective experience during the task. This questionnaire measured multiple dimensions of trust and interaction quality using items from three validated scales: NASA Task Load Index (TLX), Trust in Automation (TiA), and Human-Artifact Trust Model (HACTM). All items were measured on a 7-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree).

5 Results & Analysis

The first set of results that we have obtained are the set of pre-questions. These are a collection of 4 questions that are shown below.

The result obtained can be examined below for a sample participant and their inputted results for both groups A and B in Table 1.

The rest of the answers are shown additionally in appendix for reference.

Table 2: Sample questionnaire answers

Participant ID	The system can interpret input correctly	The system works reliably
249	6	7
823	6	7
895	6	6
827	5	6
75	6	6
711	6	7
242	3	6
616	6	7
52	7	7

Subsequently, a list of questions was provided to receive the responses of the participants after they completed their task. The result of this questionnaire for an example participant for both test groups is shown in Table 2.

Additional quantitative data that can be obtained from the rosbags includes the time taken for a movement by a participant in addition to the amount of ‘random’ moves that were made on average by participants from group A.

The average time taken for participants to move a piece for members of participant group A was 2.67 seconds and it was 3.008 for participants of group B. The way this was calculated was by first determining the mean time of movement for each participant, then averaging it for the group. The randomness used can also be examined through looking at the number of mismatched movements done by each participant from group A. Using this data we can establish the amount of random movements for each participant, what percent of the total number of moves were ‘random’ and what the mean randomness was for the entire group.

Using the data gathered from the questionnaires can then be subject to T testing. Once we have the mean value obtained for every answer in the questionnaire, we can proceed to determine the standard error and degree of freedom. Using the formula given from the analysis lecture slides, the t – value was calculated for each question. After that, a website (numiqo.com) [2] was used to find an accurate corresponding p value. Now, the data obtained and its further implications are shown in Table 3.

6 Discussion

6.1 Implications

This is a Implications subsection. From the analysis the information what can be gathered is that for the pre question segment of data gathered, the low t value and p values obtained indicate that there isn’t a significant statistical difference between the results obtained from the two groups namely A and B which were the random and non-random group respectively. What this implies is that the initial impression of participants of both groups was relatively similar.

Following the experiment and obtained questionnaire, the relevant collected and processed data for participants is indicated in the appendix. Looking at the most relevant points, which would be the

values for a significant t and thus p values, there are six questions that stand out. These would be the ones shown below as:

From these participants from group A as expected believed that their input was not being interpreted correctly. Additionally, they not only believed themselves to have not performed the task properly but above all also felt it to be unpredictable. Most of all they felt that they couldn’t trust the system as much as the control group B.

6.2 Limitations & Future Work

During the course of the study, four specific methodological limitations were identified that may have influenced the experimental data:

- (1) Specificity of Psychometric Scales: Qualitative data was derived from established scales (TiA, HCTM, and NASA-TLX). However, participant feedback indicated that certain questions lacked sufficient context for non-robotics users. For instance, the prompt "The system state was always clear to me" was occasionally misinterpreted as a question of visual clarity rather than system feedback. Similarly, composite questions asking users to rate "irritation, stress, and annoyance" simultaneously may have reduced the granularity of the emotional data collected.
- (2) Predictability of Random Noise: The experimental condition (Group A) relied on an algorithm to inject random errors into the robot’s movement. Post-hoc analysis suggests that the frequency and nature of these errors may have been pattern-like or insufficiently "alarming" to be consistently perceived as system failure, potentially dampening the contrast between Group A and Group B.
- (3) Robot Arm Dynamics (Proximity): Observations revealed that the robot arm occasionally executed sudden, large movements or lingered near the participant. Literature suggests that such sudden movements and close proximity can increase perceived risk, affecting user workload [1]. Conversely, participants generally exhibit higher comfort levels when a robot approaches them slowly and indirectly [4]. Because individual proximity data was not recorded, we could not statistically account for this variance in user comfort.
- (4) Trajectory Planning Latency: The robot utilized the MoveIt! framework with FCL for collision checking. This default checker is computationally inefficient and can be slow to calculate trajectories [6][3]. Consequently, the arm occasionally froze for several seconds while planning. Research indicates that if robot speed falls below a certain threshold, participant frustration significantly increases [5]. This latency likely acted as a confounding variable, influencing trust levels independent of the experimental condition.

Future Work To address these limitations, future studies should implement a more robust, non-deterministic randomness algorithm to ensure errors are perceived as genuine system faults. Additionally, questionnaires should be refined for clarity to suit a lay audience. Most importantly, future iterations should correlate subjective survey data with quantitative metrics—such as rosbag data (keystrokes, movement vectors) and task completion times—to accurately account for the effects of latency and proximity on user mental state.

Table 3: Statistical analysis of questionnaire responses

Questionnaire questions	Group	Mean	Std. Dev.	t-stat	p-value	Interpretation ($p < 0.05$)
The system can interpret input correctly.	A	4.73	1.90	-2.62	0.017	Significant (B higher)
The system reacts unpredictably.	A	4.27	1.42	4.13	0.001	Significant (A higher)
I trust the system.	A	4.64	1.80	-2.67	0.015	Significant (B higher)
	B	6.30	0.82			
I can rely on the system.	A	4.27	1.42	-3.08	0.006	Significant (B higher)
I think that this system performs its given role very well.	A	5.73	1.19	-2.69	0.014	Significant (B higher)
How satisfied are you with your level of performance?	A	5.73	1.10	-2.28	0.034	Significant (B higher)

7 Conclusion

The primary objective of this study was to test the hypothesis that the accuracy of robot movement directly correlates with user trust. The results obtained from the teleoperation of the Panda robot arm support this hypothesis. While statistical analysis of pre-experiment data indicated that both groups began with similar expectations, the post-experiment data revealed a significant divergence. Participants in Group A, subjected to random movement noise, reported significantly lower trust levels compared to the control Group B.

Critically, the study found that system unpredictability did not only affect confidence in the robot; it also negatively impacted the participants' self-assessment, causing them to internalize the system's errors as their own failure.

However, the study also highlighted the complexity of measuring subjective trust. We identified that factors such as the specificity of questionnaire items and the unintended predictability of the randomness algorithm may have influenced the sensitivity of our data. Furthermore, technical variables like trajectory planning delays and robot proximity were observed to potentially act as confounding variables. Therefore, while the link between accuracy and trust is evident, future work must refine the randomness algorithms and correlate subjective feedback with objective telemetry—such as reaction times and keystroke data—to provide a more granular understanding of how trust is lost and regained in human-robot interaction.

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A Research Results

- (1) Pre-Questionnaire Results - Table 4
- (2) Post-Questionnaire Results - Table 5
- (3) Workload Assessment Results - Table 6
- (4) Individual Task Performance Data Results - Table 7
- (5) Summary Task Performance by Group Results - Table 8

B Our Contribution

B.1 Mariam Machaidze

Wrote the following parts: Task Design (3), Method (4).

B.2 Amanuel Basaznew Legesse

Wrote the following parts: Results & Analysis (5), Implications (6a), determined t and p values.

B.3 Anuraag Deshpande

Wrote the following parts: Limitations & Future Work (6a), Conclusion (7), References.

B.4 Dan Akai

Wrote the following parts: Abstract, Introduction, Study Design.

centering

Table 4: Pre-Questionnaire Results

Question	Group	Mean	Variance	Std Dev	t-stat	p-value
I have already used similar systems.	A	3.55	3.27	1.81	0.51	0.615
	B	3.10	4.77	2.18		
One should be careful with unfamiliar automated systems.	A	6.00	1.40	1.18	0.48	0.636
	B	5.70	2.68	1.64		
I rather trust a system than mistrust it.	A	4.18	3.76	1.94	-0.76	0.455
	B	4.80	3.07	1.75		
Automated Systems generally work well.	A	5.00	1.40	1.18	0.78	0.444
	B	4.50	2.94	1.72		

Table 5: Post-Questionnaire Results

Question	Group	Mean	Std Dev	p-value
The system can interpret input correctly.	A	4.73	1.90	0.017*
	B	6.40	0.70	
The system works reliably.	A	5.27	1.62	0.051
	B	6.50	0.97	
A system malfunction is likely.	A	3.91	1.45	0.412
	B	4.50	1.78	
The system can take over complicated tasks.	A	4.18	2.09	0.713
	B	4.50	1.78	
The system might make unexpected errors.	A	4.55	1.63	0.399
	B	3.80	2.30	
I am confident about the system's capabilities.	A	4.73	1.42	0.254
	B	5.40	1.17	
The system's state was always clear to me.	A	5.18	2.09	0.090
	B	6.50	1.08	
The system reacts unpredictably.	A	4.27	1.42	0.001*
	B	2.00	1.05	
I was able to understand why things happened.	A	4.73	1.90	0.096
	B	6.10	1.66	
It's difficult to identify what the system will do next.	A	3.18	2.04	0.544
	B	2.60	2.27	
I trust the system.	A	4.64	1.80	0.015*
	B	6.30	0.82	
I can rely on the system.	A	4.27	1.42	0.006*
	B	6.10	1.29	
I feel I needed to be cautious while using this system.	A	4.27	1.68	0.415
	B	4.80	1.14	
I think that this system performs its given role very well.	A	5.73	1.19	0.014*
	B	6.80	0.42	
I believe that the system has all the functionalities I would expect.	A	6.09	0.83	0.199
	B	6.50	0.53	

Table 6: Workload Assessment

Question	Group	Mean	Std Dev	p-value
How mentally demanding was the task?	A	2.00	1.48	0.642
	B	2.30	1.42	
How physically demanding was the task?	A	1.64	1.29	0.907
	B	1.70	1.16	
How satisfied were you with your level of performance?	A	5.73	1.10	0.034*
	B	6.60	0.52	
How hard did you have to work to accomplish your level of performance?	A	2.36	1.36	0.519
	B	1.90	1.85	
How irritated, stressed, or annoyed were you?	A	1.82	1.25	0.671
	B	2.10	1.73	

Table 7: Individual Task Performance Data

Participant	Group	Time to Complete (s)	Avg Time Between Movements (s)
052-A	A	230.19	2.68
227-B	B	195.50	3.01
242-A	A	317.44	2.54
249-A	A	320.26	2.46
255-B	B	220.69	3.56
299-A	A	250.16	2.30
376-A	A	220.61	2.42
461-A	A	220.53	2.48
462-A	A	225.52	2.28
075-A	A	221.17	2.54
206-B	B	191.95	2.82
805-B	B	181.34	2.83
823-B	B	209.28	2.87
827-B	B	182.80	2.95
888-B	B	179.07	2.89
891-A	A	285.02	2.55
895-A	A	195.57	2.61
531-B	B	207.62	3.35
614-A	A	233.71	2.16
616-B	B	257.02	3.67
711-B	B	200.43	3.04

Table 8: Summary Task Performance by Group

Group	Avg Time to Complete (s)	Avg Time Between Movements (s)
A	247.29	2.46
B	202.57	3.10